Gravitational Lensing Lab Manual

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Measurements Using the Camera

In this section you want to take some measurements using the Nikon Z 50 II camera. Let's start with the logarithmic lens before repeating the similar steps for the linear lens.

1. Determine the radius R of the Einstein ring as seen on the paper (at a distance L from the pinhole) in the previous section. Use the formula $r/R = X/L$ to accurately determine X, the distance between the camera chip and the pinhole. r is the ring radius in pixels (meaning X is measured in units of pixels as well), which can be measured using the code in the provided Python Jupyter notebook. This serves as your calibration relating physical image size to angle. This calibration is only as long as nothing about the setup has moved.

2. Mount the camera behind the pinhole. Make sure you attach the lens mount first since the pinhole protrudes out and can damage the camera. Connect the camera to the computer and open the NX Tether application. Make sure the camera shows up as the connected camera. Every time you take a picture, the photo should save directly to the computer.

3. Align to get the Einstein ring and move the camera so that the ring is centered on the screen.

4. Take a picture of the Einstein ring.

5. Turn off the light source or use a book to block the light source and take a picture that will be used later for background subtraction.

6. Change the values of D_d, D_s, D_{ds} measure them, and take more pictures. Measure the ring radius r and use the values of $D_{sc} = X$ from step (1). Calculate $D = \frac{D_d D_s}{D_{ds}}$, $\theta_E = \arctan \frac{r}{D_{sc}}$ and verify that $\theta_E \propto D^{-1/2}$.

7. Use the Jupyter notebook to measure the ring radius r.

8. Choose one set of values of D_d, D_s, and D_{ds}. Move the light source from the axis a distance η and take a picture and repeat for several distances of η. Make sure you measure the η values you use.

9. Use the Jupyter notebook to measure δ_+ and δ_-, the distances between the two images and the center of the Einstein ring.

10. Calculate $\theta_E = \arctan \frac{r}{D_{sc}}$, $\beta = \arctan \frac{\eta}{D_s}$, $\theta_+ = \arctan \frac{\delta_+}{D_{sc}}$, $\theta_- = \arctan \frac{\delta_-}{D_{sc}}$. Verify that $\theta_\pm = \frac{1}{2}(\beta \pm \sqrt{\beta^2 + 4\theta_E^2})$

11. Repeat this 5 times for different positions of the light source and lens, to map out the dependence $\theta_\pm$ on the dimensional parameters. You will want to produce graphs similar to those shown in the *Basic Knowledge* document.

Now repeat the same steps for the linear lens. Remember that in this case, if $\beta < \theta_E$, $\theta\pm = \beta \pm \theta_E$ and if $\beta > \theta_E$, there is only one image at $\theta = \theta_+ = \beta + \theta_E$.

Intensity Measurement

1. Use the Jupyter notebook to perform background subtraction.
2. Then, use the Jupyter notebook to measure the intensity of the two images and determine the ratio I_+/I_-.
3. Compare the measured intensity ratios with the theoretical ratios μ_+/μ_- where $\mu_\pm = \frac{u^2+2}{2u\sqrt{u^2+4}} \pm \frac{1}{2}$, with $u = \beta\theta_E^{-1}$ for the logarithmic lens and $u_\pm = \frac{\theta_\pm}{\beta} = 1 \pm \frac{\theta_E}{\beta}$ for the linear lens.